

C++ Development Commands on Linux

1. Basic C++ Compilation and Execution Commands

`g++ file.cpp -o output` - Compile a C++ file using `g++`

`./output` - Run the compiled program

`g++ file1.cpp file2.cpp -o output` - Compile multiple C++ files

`g++ -Wall file.cpp -o output` - Compile with all warnings

`g++ -std=c++11 file.cpp -o output` - Compile using C++11 standard

`g++ -g file.cpp -o output` - Compile with debug symbols

`g++ -O2 file.cpp -o output` - Compile with optimizations

2. File Management Commands (Linux)

`ls` - List files in the current directory

`cd directory_name` - Change directory

`pwd` - Show current directory path

`mkdir folder_name` - Create a new folder

`rm file.cpp` - Delete a file

`rm -r folder_name` - Delete a directory recursively

`cp file1.cpp file2.cpp` - Copy files

C++ Development Commands on Linux

`mv old.cpp new.cpp` - Rename or move files

`touch file.cpp` - Create a new empty file

3. File Viewing and Editing

`cat file.cpp` - Display contents of file

`less file.cpp` - View file with scroll

`nano file.cpp` - Open file with Nano editor

`vim file.cpp` - Open file with Vim editor

`code file.cpp` - Open with VS Code (if installed)

4. Debugging C++ Code

`g++ -g file.cpp -o output` - Compile with debugging info

`gdb ./output` - Start GNU Debugger

`break main` - Set breakpoint at `main()`

`run` - Run the program in GDB

`next` - Execute next line (step over)

`step` - Step into function

`print var` - Print value of variable

C++ Development Commands on Linux

quit - Exit GDB

5. Makefiles

make - Build project using Makefile

make clean - Remove compiled files

make target_name - Build specific target from Makefile

6. Package Installation (Compiler, Build Tools)

sudo apt update - Update package list (Debian/Ubuntu)

sudo apt install g++ - Install C++ compiler

sudo apt install build-essential - Install essential build tools

sudo yum install gcc-c++ - Install g++ on RHEL/CentOS

which g++ - Check g++ location

g++ --version - Check g++ version

7. Searching and Grepping

grep "main" file.cpp - Find "main" in file

find . -name "*.cpp" - Find all C++ files

grep -rnw . -e "functionName" - Recursive search in all files

8. Archiving and Compression

C++ Development Commands on Linux

`tar -cvf archive.tar folder/` - Create tar archive

`tar -xvf archive.tar` - Extract tar archive

`zip -r file.zip folder/` - Create zip archive

`unzip file.zip` - Extract zip archive

9. Permissions

`chmod +x output` - Make file executable

`chmod 755 script.sh` - Set permission for script

`chown user:user file.cpp` - Change ownership

10. Running in Background / Redirecting Output

`./output &` - Run in background

`./output > output.txt` - Redirect output to file

`./output >> log.txt` - Append output to file

`./output < input.txt` - Provide input from file